

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester Bachelor of Technology (B.Tech) (CE/IT)

University Examination May 2011

CE 201 Data Structures & Algorithms

Date: 17/05/2011

Time: 10:00 a.m to 1:00 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1 Answer the following questions.

- a) What is Data structure? Explain different data structure giving suitable examples. 02
- b) Define recursion and write the applications of recursion. 02
- c) What is a sparse matrix? Explain with example.
- d) What is Deque? Explain in brief. 02
- e) State True or False with justification: 02
 - (i) A procedure working on a circular linked list may go into an infinite loop. 03
 - (ii) A priority Queue is implemented using stack.
 - (iii) Stacks are used in recursion.

Q.2 Answer the following questions.

- a) Explain how circular queue is advantageous as compared to simple queue? Write an algorithm to insert and delete an element in a circular queue. 04
- b) Explain collision in Hashing. Explain any one-collision resolution technique in detail. 03
- c) Trace the algorithm for the converting given infix expression to postfix expression: $((A + ((B^C) - D)) * (E - (A/C)))$. 05

OR

Q.2 Answer the following questions

- a) Describe Triangular and Sparse arrays. How such arrays get generated? What are the options for mapping of these arrays? 04
- b) Explain advantages and disadvantages of Indexed sequential files. 03
- c) Write an algorithm to print factorial of a number using stack. 05

Q.3

- a) Describe Tower of Hanoi Problem and give its three rules. Also give Recursive algorithm for Tower of Hanoi Problem. 06
- b) Write an algorithm to reverse the given linked list. 06

OR

Q.3 Answer the following questions.

- a) What is circular linked list? Explain the benefits and drawbacks of circular linked list. 06
- b) Generate index for each data using hash function and arrange them into an array A[10]. Use linear probing to resolve collision. 06
Data: 234, 806, 188, 697, 187, 835, 533, 107
Hash function $H(x) = (x \bmod 10) + 2$.

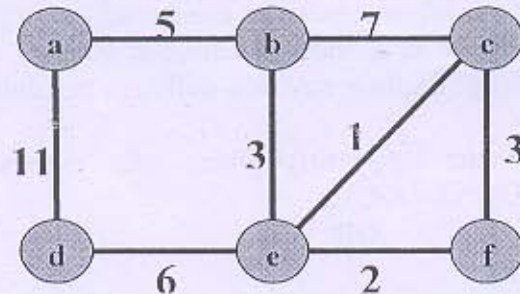
SECTION – II

- Q.4 Answer the following questions.**
- | | | |
|----|---|----|
| a) | Explain following terms.
1. Path 2. Out degree 3. Complete binary tree 4. Cycle | 04 |
| b) | Explain Binary search technique with example. | 02 |
| c) | Write an algorithm of Bubble Sort. | 02 |
| d) | State True or False with justification: | 03 |
| | (i) AVL trees are weight-balanced trees, whereas Binary trees are height-balanced trees. | |
| | (ii) Breath first traversal of a graph proceeds level by level whereas depth-first traversal follows first a path 'from starting node to an ending node, then another path from the start to an end and so forth. | |
| | (iii) Every tree is a graph. | |

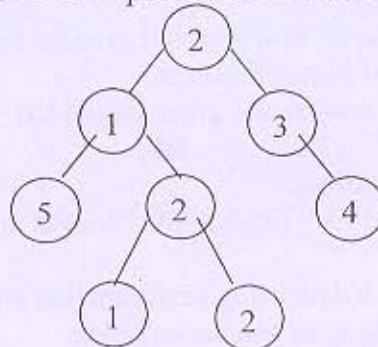
- Q.5 Answer the following questions.**
- | | | |
|----|---|----|
| a) | Show the trace of selection sort on following. 33, 9, 65, 82, 75, 88, 92, 4, 12, 41 | 04 |
| b) | Insert the following values into a binary search tree in the following order: 1, 3, 5, 8, 0, 9, 12, 4, 2, and 15. | 04 |
| c) | Explain threaded binary tree and give node representation in threaded binary tree. | 04 |

OR

- Q.5 Answer the following questions.**
- | | | |
|----|---|----|
| a) | Write and explain an algorithm for Depth First Search (DFS) traversal for undirected graph with a suitable example. | 04 |
| b) | Explain storage representation techniques Adjacency Matrix in graph. | 02 |
| c) | Find the Minimum Spanning Tree using Krushkal's Algorithm starting from node a. | 06 |



- Q.6 Answer the following questions.**
- | | | |
|----|---|----|
| a) | Write an algorithm for radix sort and sort the following data using radix sort. 50, 31, 66, 309, 408, 201, 49, 53, 576, 11. | 06 |
| b) | Write the inorder, Preorder and post order traversal of following tree: | 06 |



OR

- Q.6 Answer the following questions.**
- | | | |
|----|---|----|
| a) | What is AVL tree? Discuss Single and double rotation in AVL tree. | 06 |
| b) | Create min-heap for the keys 23, 56, 4, 12, 58, 18, 88 using successive inserts (starting from 23). | 06 |